

Time Conversion in ArchesWeather

In this notebook, I show that something is most likely going wrong with the time conversion in ArchesWeather.

Specifically, when passing a weather state carrying a timestamp in seconds since 1970, the generative model fails to convert it correctly to a datetime. As a result, the hour of the day is extracted incorrectly.

```
In [ ]: import marimo as mo
import torch
import numpy as np
import pandas as pd
from datetime import datetime
```

```
In [ ]: from geosarches.dataloaders.era5 import Era5Forecast
```

```
In [ ]: from src.utils import get_dataset
```

```
In [ ]: def tensor_timestamp_to_string(
    timestamp: torch.Tensor,
    fmt: str = "%Y-%m-%d %H:%M:%S",
) -> str:
    ts = timestamp.item()
    return datetime.utcfromtimestamp(ts).strftime(fmt)
```

```
In [ ]: ds = get_dataset()
```

```
0it [00:00, ?it/s]
1it [00:00, 4.26it/s]
4it [00:00, 15.14it/s]
```

First five timestamps in the dataset:

```
In [ ]: all_timestamps = [ts[2] for ts in ds.timestamps]
all_timestamps[0:5]
```

Select a weather state

I now select the weather state from the `Era5Forecast` object corresponding to the following timestamp:

```
In [ ]: x_start = ds[2]
```

second state and corresponding timestamp:

```
In [ ]: tensor_timestamp_to_string(x_start["timestamp"])
```

```
'2020-01-02 12:00:00'
```

Deterministic Model

To make a prediction, we pass the weather state to the deterministic model. The model converts the timestamp as follows:

```
time = pd.to_datetime(x_start["timestamp"].cpu().numpy(),  
unit="s").tz_localize(None)
```

```
print(time, torch.tensor(times_det.month),  
torch.tensor(times_det.hour))
```

```
In [ ]: # before embedding time in ForecastModuleWithCond  
times_det = pd.to_datetime(x_start["timestamp"].cpu().numpy(), unit="s").tz_  
(f"timestamp: {times_det}, month: {torch.tensor(times_det.month)}, hour: {tc
```

```
'timestamp: 2020-01-02 12:00:00, month: 1, hour: 12'
```

The conversion works as expected.

Generative Model

We pass the same state to the generative model, which converts the timestamp as follows:

```
time = pd.to_datetime(x_start["timestamp"].cpu().numpy() *  
10**9).tz_localize(None)
```

```
print(time, torch.tensor(times_det.month),  
torch.tensor(times_det.hour))
```

```
In [ ]: # in the sampling procedure before embedding time  
times = pd.to_datetime(x_start["timestamp"].cpu().numpy() * 10**9).tz_locali  
(f"timestamp: {times}, month: {torch.tensor(times.month)}, hour: {torch.tens
```

```
'timestamp: 1969-12-31 23:59:58.179704832, month: 12, hour: 23'
```

The conversion is incorrect.

As a consequence, the generative model cannot properly use the hour of the day as conditioning information. The same issue also affects the month, except for timestamps in January.